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1 August 2022

Software Engineering

SC2006/CE2006/CZ2006

Introduction

Outline of Lecture #1

1. **What is Software Engineering?**
2. **Software Engineering Activities**
3. **Course Logistics**

1. What is Software Engineering?

Software Engineering is:

The production of maintainable, fault-free software that meets the user's requirements and is delivered on time and within budget.

2. Software Engineering Activities

Is Software Engineering just Coding?

What are the possible activities in engineering a software product?

What were the engineering activities needed to achieve this?



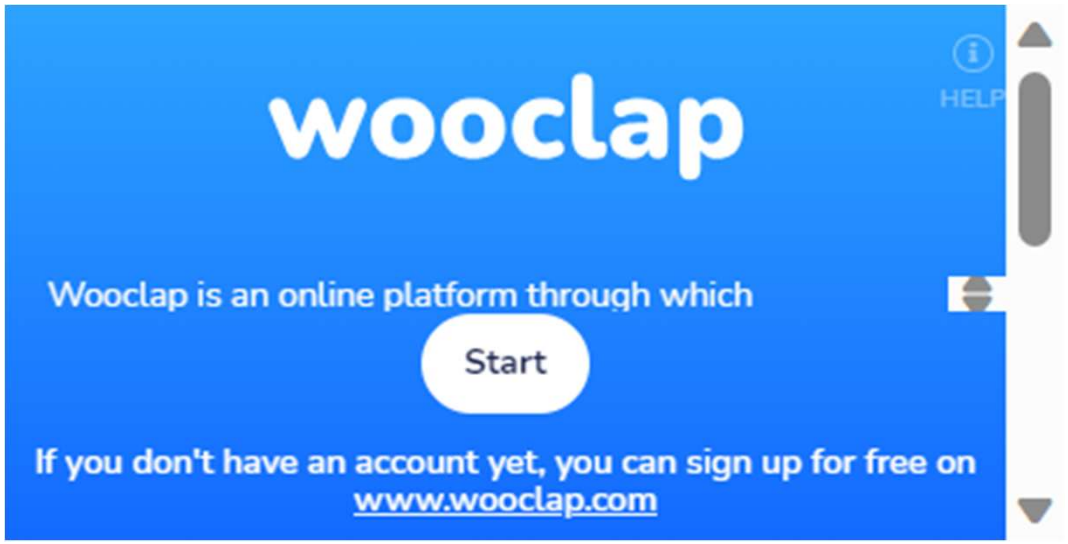






Analogy with Construction Engineering





Deciding the purpose of the building.

Deciding how many room, how big are the rooms, what the building will look like……

Designing the building. Deciding on the materials for construction, the power, plumbing and other services….

Project planning. Scheduling. Teamwork.

Simulations and testing.

Construction.

Operation and Maintenance.

All engineering requires the same activities

1. **Requirements elicitation** – specify what are we going to make, what it will do and how it will do it.
2. **Design it** – What will it look like and how will we make it?
3. **Build it** – to the design and requirements specification.
4. **Test it** – to ensure it meets the requirements and is error free.
5. **Release it** and **maintain it**.

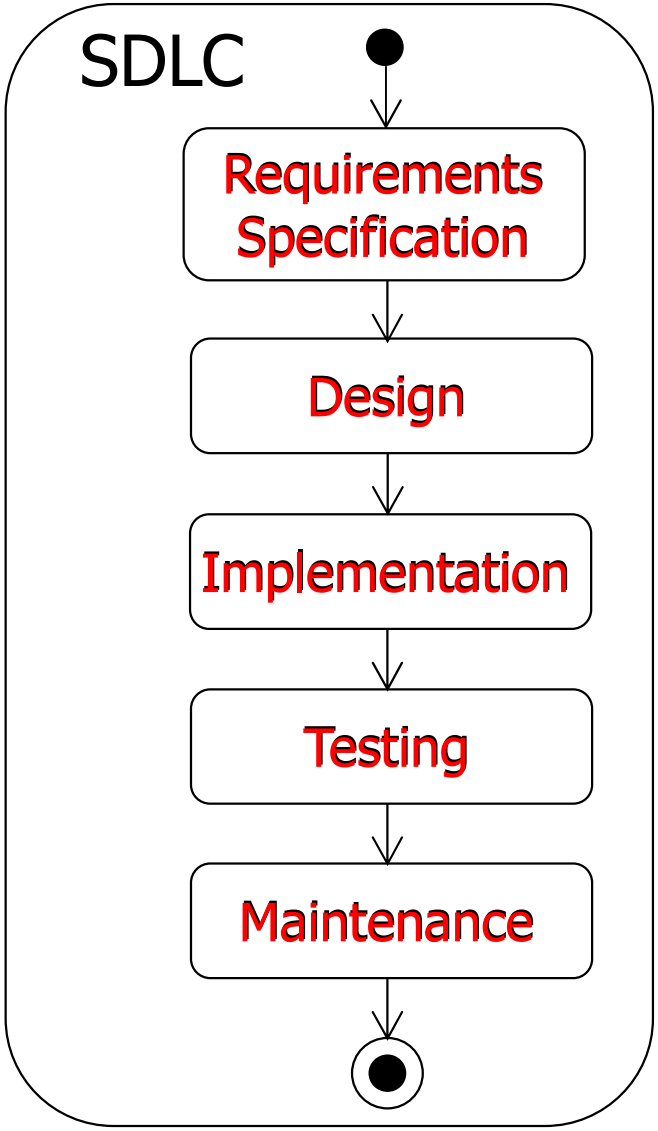






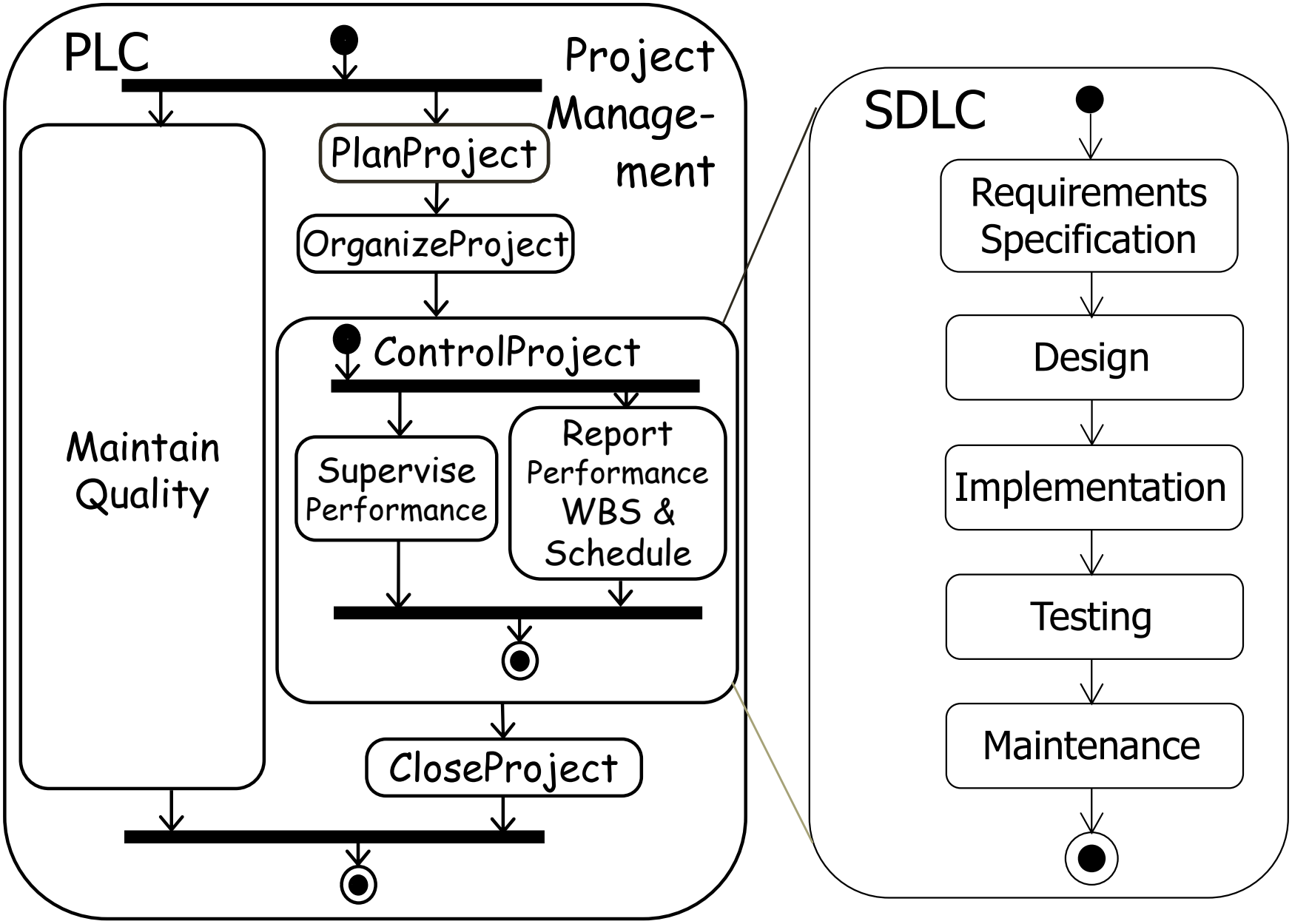


Software Engineering Activities



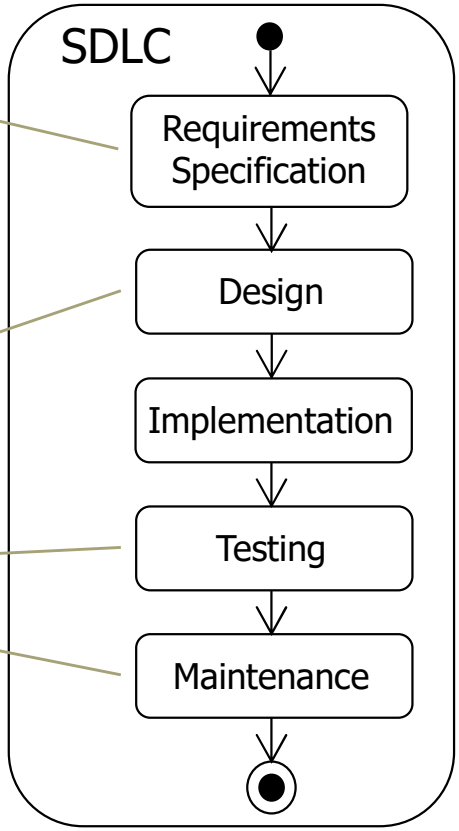
Work Products	Milestones
•Software Requirement Specification •Prototype	•Requirements Review
•Software Design Document •Interface Design Document •Test Cases •Models	•Preliminary & Final Design Reviews
•Source Code •System Build •Technical Documentation •User Manual •Test Report	•User Acceptance Test •General Availability Release
•Change Request •Bug Report	•Patch Release •Upgrade

Software Engineering Activities



Main Topics to be covered

- Requirements Elicitation and Analysis
- Software Process Models
- Agile Software Process
- System and Object Design
- Testing and Maintenance







Lectures for large and medium projects



Practices in lab project

3. Course Logistics and Mechanics

- 20+ lectures
- 9 tutorials (5%)
- 1 group project (45%) With 5 x 2 hours lab sessions
- Exam (50%)

Where learning happens!

We want Active Learning: Do more than just listening!

Tutorials

start in Week3 !

- Problem-based learning through Grain Elevator System case study and other questions/problems
  - All tutorials are released on NTULearn during week 1.

- You Should
  - Come to each tutorial prepared to present and discuss
  - Do NOT wait for the tutors answers (most questions do not have model answers as there is often not a single correct answer!)

Group Project (50%)

start in Weeks 2 and 3 !

- The project problem and five lab manuals is available on NTULearn during week 1

- You Must

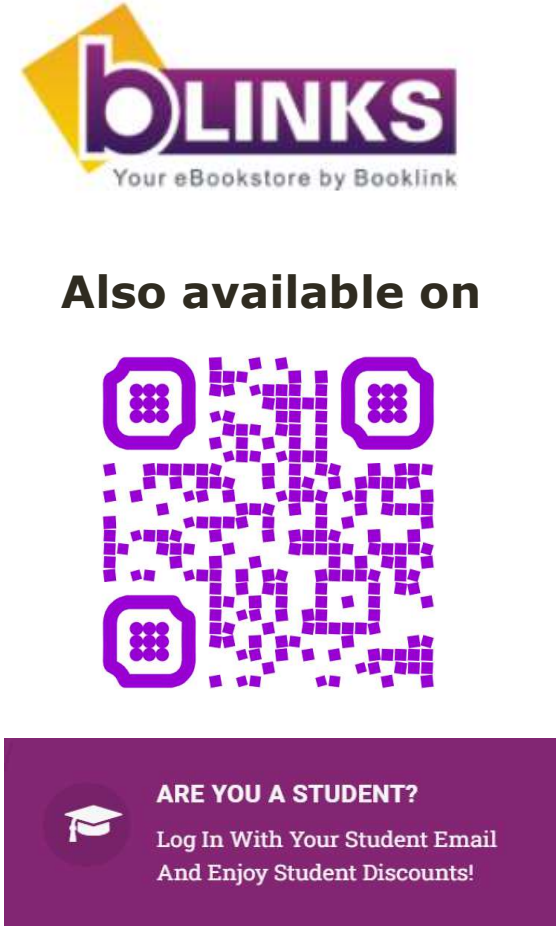
SWE lab: check your timetable

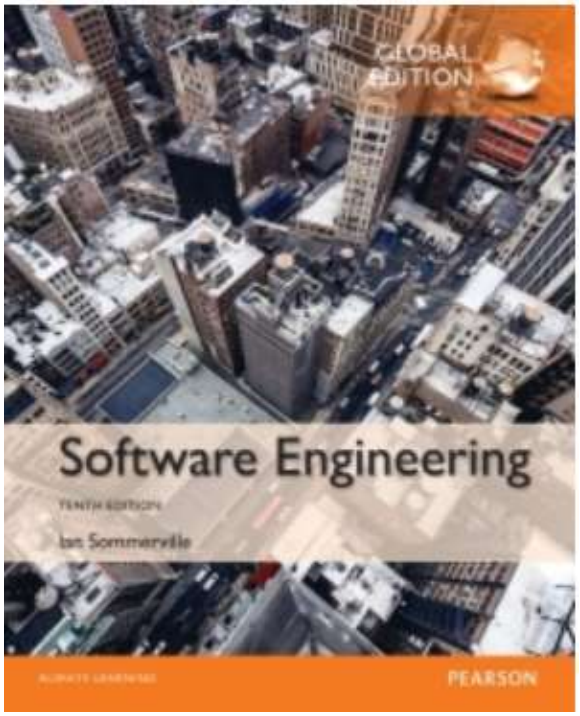
  - Work in a team (4-6 members)
  - Plan your work, start early, proceed iteratively
  - Contribute to all key Software Engineering activities
  - Meet deliverables for labs #1, #2, #3 (5% for each lab)
  - Demonstrate your working product in Lab#5 (15%)
  - Deliver documentation at end of the semester (15%)

Course Reference Books

- **Ian Sommerville, Software Engineering, 10th Edition, ISBN 10: 1-292-09613-6, Pearson, 2017.**
- **Bernd Breugge, Allen H Dutoit, *Object-Oriented Software Engineering: using UML, patterns and Java*, 3rd Edition, ISBN 10: 0-13-815221-7, Pearson, 2010.**
- **Martin Fowler, UML Distilled: a brief guide to the standard object modelling language, 3rd Edition, ISBN 0-321-19368-7, Pearson, 2004.**
- **Christopher Fox, *Introduction to Software Engineering Design: Process, Principles, and Patterns with UML2*, 1st edition, Pearson Education/Addison-Wesley, 2006.**
- **Lee Copeland, *A Practitioner's Guide to Software Test Design*, ISBN 9781580537919, Artech House, 2004.**
- **Thomas B. Hilburn, Massood Towhidnejad, Software Engineering Practice A Case Study Approach, 1st Edition, ISBN 9780367688646, Chapman & Hall, 2020.**







Software Engineering 10e by Sommerville

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